CAPSTONE PROJECT REPORT

Report 1 – Project Introduction

– Can Tho, May 2025 –

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I. Project Report

1. Status Report

#	Work Item	Status	Notes (Work Item in Details)
1		Pending	
2		In Progress	
3		Completed	

2. Team Involvements

#	Task	Member	Notes (Task Details, etc.)
1		KienNT	
2		TuanTV	
3		AnhLM	

3. Issues/Suggestions

#	Issue	Status	Notes (Solution, Suggestion, etc.)
1		Pending	
2		In Progress	
3		Completed	

II. Project Introduction

1. Overview

1.1 Project Information

- Project name: **S.M.I.L.E: Smart Dental Management and Diagnosis System**
- Project code: **SMILE**
- Group name: **SEP490.7**
- Software type: **Web Application**

1.2 Project Team

a. Supervisor

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b. Team Members

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2. Product Background

Patients seeking dental care in major Vietnamese urban centers currently navigate a fragmented and opaque journey. On average, a new patient spends approximately 45 to 60 minutes just searching for a reliable clinic, often relying on unverified social media reviews and static websites that lack clear information on practitioner credentials or actual treatment costs. Once a clinic is selected, the booking process is predominantly manual—conducted via phone or basic messaging apps—resulting in frequent scheduling conflicts and a "no-show" rate that can reach up to 25% for some practices.

During the consultation phase, patients often experience high levels of "dental anxiety" because practitioners lack the digital tools to visually demonstrate complex treatment outcomes, such as multi-stage implant procedures or orthodontic shifts. Furthermore, clinical records are frequently stored in disconnected local databases or physical files, making it nearly impossible for patients to maintain a comprehensive, portable medical history or for clinics to coordinate multidisciplinary care effectively.

From an operational perspective, clinic owners struggle with significant inefficiencies. Expensive dental chair time is often underutilized due to the lack of an intelligent scheduling system that can predict procedure durations or fill last-minute cancellations. Additionally, clinics face high administrative overhead as staff spend hours manually verifying insurance details and sending appointment reminders. Post-treatment, the lack of an integrated follow-up system leads to a "care gap," where patients often neglect preventive maintenance, ultimately undermining the long-term success of high-value treatments like All-on-X rehabilitations. These systemic issues contribute to a lack of trust and operational waste across the domestic dental healthcare sector.

2.1 Frontend Technologies

2.1.1 NEXTJS

Next.js is an open-source web development framework created by Vercel enabling React-based web applications with server-side rendering and generating static websites. It enhances the standard React environment by providing out-of-the-box solutions for routing, optimization, and data fetching, allowing developers to build fast, SEO-friendly, and user-friendly full-stack applications.

Figure 1. NextJS[1]

Advantages

- **SEO Performance:** Pre-rendering content on the server makes it easily indexable by search engine crawlers.
- **Fast Load Times:** Features like code splitting and prefetching ensure only the necessary code is loaded for each page.
- **Developer Experience (DX):** Fast Refresh, zero-config setup, and TypeScript support out of the box.
- **Deployment:** Highly optimized for Vercel but can be containerized using Docker for any hosting provider.
- **Full-stack Capability:** Eliminates the need for a separate backend for simple logic via Server Actions and API routes.

2.2 Backend Technologies

2.2.1 NestJS

[1] https://nextjs.org/

NestJS is a progressive, open-source Node.js framework designed for building efficient, reliable, and scalable server-side applications. It leverages modern JavaScript, is built with TypeScript (ensuring strict type safety), and combines elements of Object-Oriented Programming (OOP), Functional Programming (FP), and Functional Reactive Programming (FRP). Under the hood, NestJS makes use of robust HTTP server frameworks like Express by default, while optionally providing configuration to use Fastify for maximum performance.

Figure 2. NestJS[23]

Advantages

- **Standardized Architecture:** Provides an out-of-the-box, highly structured modular architecture, enabling large development teams to maintain, scale, and organize codebases effortlessly and consistently.
- **Superb TypeScript Support:** Built natively with TypeScript, offering the full benefits of static typing, advanced IntelliSense, and early detection of bugs during compile time.
- **Powerful Ecosystem:** Inherits the massive npm ecosystem while providing first-class, built-in support for cutting-edge technologies like GraphQL, WebSockets, Microservices, and popular ORMs (TypeORM, Prisma).
- **Dependency Injection (DI):** Features a powerful Inversion of Control (IoC) container that automates dependency management, making applications highly decoupled and exceptionally easy to unit test.
- **Enterprise-Ready:** Heavily inspired by the architectural principles of Angular and Spring Boot, making it the premier choice for building scalable microservices and heavy enterprise backends.

2.2.2 Gin Gonic

Gin is a high-performance HTTP web framework written in Go (Golang). It features a Martini-like API but with performance up to 40 times faster, thanks to **httprouter**. Gin allows developers to build robust, scalable RESTful APIs and microservices with minimal memory footprint and maximum execution speed.

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[3] https://nestjs.com/

Figure 3. Gin-Gonic Framework

Advantages

- **Speed:** One of the fastest web frameworks in the Go ecosystem, ideal for high-traffic applications.
- **Low Memory Usage:** Optimized for low resource consumption, making it perfect for microservices and serverless environments.
- **Productivity:** Striking a balance between the minimalism of the standard library and the features of a full framework.
- **Stability:** A very mature ecosystem with extensive documentation and a large community for troubleshooting.
- **Easy Integration:** Seamlessly works with standard Go net/http packages and libraries.

2.3 Database

2.3.1 PostgreSQL

PostgreSQL is a powerful, open-source object-relational database system (ORDBMS) with over 30 years of active development that has earned it a strong reputation for reliability, feature robustness, and performance. It extends the SQL language combined with many features that safely store and scale the most complicated data workloads.

Figure 4. PostgreSQL[4]

Advantages

- **Reliability and Stability:** Known for being rock-solid and preventing data corruption.
- **Open Source:** Free to use, with no licensing costs and a vibrant community.
- **Complex Queries:** Excellent support for complex queries, joins, and foreign keys.
- **Scalability:** Capable of handling large volumes of data and high concurrent user loads.
- **Geospatial Support:** With the PostGIS extension, it is the industry standard for geospatial data.

2.3.2 Redis

Redis (Remote Dictionary Server) is an open-source, in-memory data structure store, used as a database, cache, and message broker. It supports various data structures such as strings, hashes, lists, sets, and more. Because it resides in memory, Redis delivers sub-millisecond response times, making it ideal for real-time applications.

Figure 7. Redis[5]

Advantages

- **Exceptional Performance:** Extremely fast, capable of handling millions of requests per second.
- **Flexible Data Structures:** Not limited to simple strings; supports lists, sets, maps, etc.
- **Simple Usage:** Easy to install, configure, and use.
- **Versatility:** Can be used for caching, session management, real-time analytics, and pub/sub messaging.
- **Community:** Large community and support for almost every programming language.

2.4 AI and Machine Learning

2.4.1 Python

Python is a high-level, interpreted, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python is dynamically typed and

[4] https://medium.com/codex/intro-to-postgresql-c8da31335c34

[5] https://redis.io/

garbage-collected, supporting multiple programming paradigms, including structured (particularly procedural), object-oriented, and functional programming. It is often described as a "batteries included" language due to its comprehensive standard library.

Figure 8. Python[6]

Advantages

- **Beginner-Friendly:** Its simple syntax closely resembles English, making it easy to learn and share.
- **High Productivity:** Allows developers to write less code to achieve more compared to languages like C++ or Java.
- **Massive Ecosystem:** Provides powerful libraries for AI/ML (TensorFlow, PyTorch), Data Science (Pandas, NumPy), and Web (Django, Flask).
- **Integration Capabilities:** Easily integrates with C, C++, and Java (via Jython), making it an excellent "glue language."
- **Large Community:** Extensive support, tutorials, and third-party packages available through PyPI (Python Package Index).

2.4.3 UNET

U-Net is a convolutional neural network architecture developed primarily for biomedical image segmentation. Its name originates from its symmetric, U-shaped structure, which consists of a contracting path (encoder) and an expansive path (decoder). Unlike standard CNNs used for classification that reduce an image to a single label, U-Net is designed for pixel-wise prediction, meaning it assigns a class to every single pixel in the input image. This architecture is particularly famous for its ability to work effectively with very few training images while producing highly precise segmentation masks.

[6] https://www.python.org/

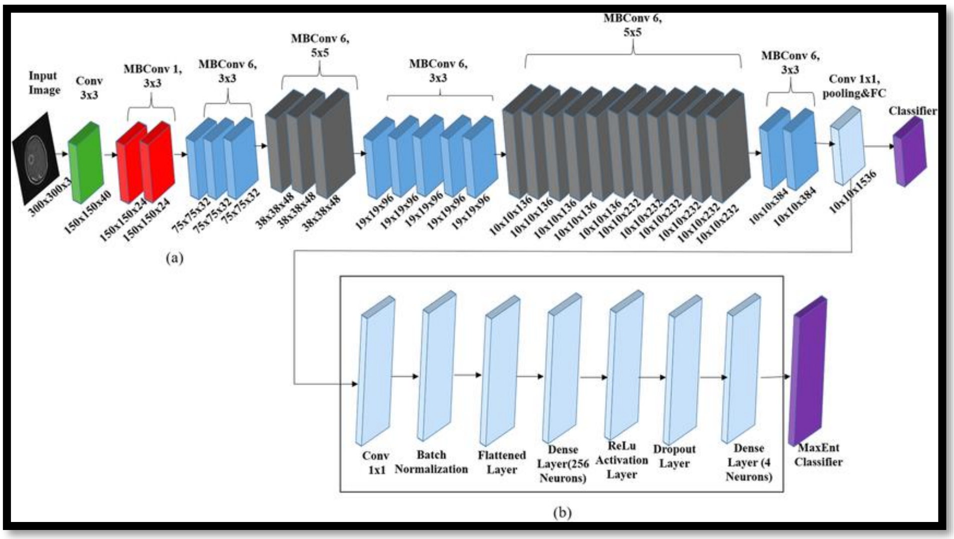

Figure 10. efficientnet[7]

Advantages

- **Data Efficiency:** Performs exceptionally well even with a **limited dataset**, which is crucial in medical fields where labeled data is scarce.
- **Spatial Precision:** Skip connections recover spatial information lost during downsampling, leading to very sharp and accurate boundaries.
- **High Resolution Output:** Generates a segmentation mask that matches the resolution of the input image.
- **Versatility:** Although born in medicine, it is now a baseline for satellite imagery, autonomous driving, and even generative AI (Diffusion Models).
- **Speed:** Inference is relatively fast once trained, making it suitable for many clinical and industrial applications.

2.5 Deployment and Testing

2.5.1 Docker

Docker is an open-source platform that automates the deployment, scaling, and management of applications using containerization. It allows developers to package an application with all of its dependencies (libraries, runtime, system tools, etc.) into a standardized unit called a container. This ensures that the application runs consistently regardless of the environment, solving the "it works on my machine" problem.

[7] https://www.geeksforgeeks.org/machine-learning/u-net-architecture-explained/

Figure 13. Docker[8]

Advantages

- **Consistency:** Ensures the application runs the same way in development, testing, and production.
- **Efficiency:** Uses fewer resources than traditional virtual machines (VMs) because containers share the OS kernel.
- **Rapid Deployment:** Containers can be spun up or down in seconds, facilitating rapid scaling and deployment.
- **Microservices Support:** Ideal for microservices architecture where each service runs in its own container.
- **DevOps Integration:** Plays a crucial role in modern CI/CD pipelines.

[8] https://www.docker.com/

3. Existing Systems

3.1 Related works

3.1.1 An interpretable deep learning framework for multi-class dental disease classification from intraoral RGB images

This study presents a deep learning framework specifically designed for classifying multiple dental diseases from intraoral RGB photographs. The research addresses the challenge of automated dental diagnosis by developing an interpretable AI model that can identify various oral pathologies from standard clinical images. The framework emphasizes explainability in AI decision-making, which is crucial for clinical acceptance and trust. The multi-class classification approach aligns directly with our AI-Powered Diagnostic Assistance feature, providing foundation for implementing disease detection capabilities in dental practice management systems.

Ali, D.A., Sadeeq, H.T.: An interpretable deep learning framework for multi-class dental disease classification from intraoral RGB images. Statistics, Optimization and Information Computing 14, 3380–3397 (2025)

3.1.2. Deep learning for oral health: benchmarking ViT, DeiT, BEiT, ConvNeXt, and Swin Transformer

This comprehensive study evaluates the performance of various state-of-the-art transformer architectures for oral health applications. The research benchmarks Vision Transformer (ViT), Data-efficient Image Transformer (DeiT), Bidirectional Encoder representation from Image Transformers (BEiT), ConvNeXt, and Swin Transformer models on dental image analysis tasks. The comparative analysis provides valuable insights into selecting optimal AI architectures for dental diagnostic systems. This work directly supports our technical decisions for implementing the Dental Image Service and Oral Check Service components, offering evidence-based model selection for achieving high accuracy in automated dental disease detection.

George, A.B., Bathini, S., Niranjana, S.R.: Deep learning for oral health: benchmarking ViT, DeiT, BEiT, ConvNeXt, and Swin Transformer. arXiv preprint arXiv:2509.23100v1 (2025)

3.1.3. Two-step pipeline for oral diseases detection and classification: a deep learning approach

This research proposes a systematic two-step pipeline approach for automated oral disease detection and classification using deep learning methodologies. The study establishes a workflow that first detects the presence of oral pathologies and then classifies them into specific disease categories. The pipeline architecture addresses practical implementation challenges in clinical settings, including image preprocessing, feature extraction, and classification stages. This approach provides a foundational framework for our Medical Image Management system, particularly informing the design of our image upload, AI analysis, and reporting workflow that enables seamless integration of AI diagnostic capabilities into dental practice management.

Araújo, A.L.D., et al.: Two-step pipeline for oral diseases detection and classification: a deep learning approach. Frontiers in Oral Health 6, 1659323 (2025)

3.2 Existing systems

3.2.1 Lan Anh Dental[9]

Lan Anh Dental is a long-standing and prestigious dental system in Vietnam, positioned as a provider of comprehensive oral healthcare for families. The platform focuses on building credibility through a team of highly qualified specialists and modern infrastructure, catering to diverse needs from general dentistry to advanced aesthetics.

Key Features:

- Comprehensive Service Portfolio: Offers a wide range of services including orthodontics, porcelain crowns, and pediatric dentistry.
- Doctor Profiles: Detailed information regarding the professional qualifications and experience of the medical team, enhancing user trust.
- Educational Blog: A system of in-depth articles on oral health care, supporting patient education and engagement.
- Direct Booking Interface: Integrated tools for quick online consultation registration and appointment scheduling.

Advantages:

- Established Reputation: A long-term brand with a vast and loyal customer base.
- Family-Oriented Approach: The environment and services are optimized for patients of all ages within a family unit.
- Professional Content: The website features professional medical content and authentic imagery from clinical practice.

Disadvantages:

- Traditional UI/UX: The website interface remains somewhat conventional and lacks a modern, high-impact visual experience.
- Information Overload: The large volume of content may make it difficult for users to navigate quickly to specific information.

[9] nhakhoalananh.com

3.2.2 Vietnam Implants [10]

Vietnam Implants is an online platform specializing in advanced dental implant techniques and high-tech dentistry, targeting high-end domestic clients and international dental tourists. The website is optimized to demonstrate clinical expertise that meets global standards.

Figure 1. Viet Nam Implants

Key Features:

- Specialized Implant Focus: Deep focus on complex grafting and implant techniques such as All-on-4 and All-on-6.
- International Patient Support: Integrated information regarding logistics, including accommodation and transportation for overseas patients.
- Before & After Gallery: A visual library of treatment results serving as empirical evidence of clinical capability.
- Multi-language Support: Multilingual functionality to effectively serve a global clientele.

Advantages:

- High-Tech Positioning: Establishes a leadership position in the most advanced dental technologies.
- Global Standards: Clinical processes and quality are certified by international dental organizations.
- Niche Market Appeal: Effectively attracts target audiences with specific high-budget requirements.

Disadvantages:

- High Cost Perception: The focus on high-tech specialized techniques may deter general-interest patients due to perceived costs.
- Narrow Service Focus: Users seeking basic dental services may find the platform’s focus too exclusive.

[10] vietnamimplants.com

3.2.3 Hospital of Odonto-Stomatology HCM City[11]

The Hospital of Odonto-Stomatology HCM City is a leading public medical institution specializing in odonto-stomatology and maxillofacial surgery. As a specialized hospital under the Department of Health, its platform serves as a formal gateway for public healthcare services, focusing on clinical excellence, emergency dental care, and specialized surgical interventions that go beyond the scope of private clinics.

Website:

BỆNH VIỆN RĂNG HÀM MẶT TPHCM
NỤ CƯỜI CỦA BẠN, TRÁCH NHIỆM CỦA CHÚNG TÔI

Key Features:

- Public Health Information Portal: Provides official guidelines on dental health, social insurance (BHYT) policies, and administrative procedures.
- Specialized Department Directory: Detailed sections for specialized units such as Maxillofacial Surgery, Orthodontics, and Implantology.
- Online Appointment Booking: A dedicated interface for patients to schedule consultations, aimed at reducing on-site waiting times.
- Academic & Research Repository: Contains information regarding medical training programs, scientific research, and international cooperation activities.

Advantages:

- Institutional Authority: As a public hospital, it possesses the highest level of clinical trust and legal accreditation in the region.
- Complex Case Management: Equipped to handle severe pathologies and emergency surgeries that private practices typically refer out.
- Affordability & Insurance: Offers competitive pricing and supports government health insurance, making high-quality care accessible to the general public.

Disadvantages:

- Institutional UX Design: The website follows a traditional governmental layout, which may feel less "vibrant" or user-centric compared to private dental brands.
- High Patient Volume: The digital interface reflects a high-traffic environment, which can sometimes lead to slower response times for online inquiries.
- Limited Aesthetic Marketing: Unlike private competitors, the platform focuses more on clinical functions than on lifestyle-oriented aesthetic marketing.

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[11] bvranghammat.com

3.2.4 Others

Beyond the major dental hubs, a significant portion of the market utilizes generic clinical management software or localized legacy systems. These platforms are designed primarily for backend administrative tasks such as basic billing, inventory logging, and demographic data entry. While they provide a digital alternative to paper records, they lack the advanced AI-driven diagnostics integrated ecosystems.

Key Features:

- Basic Electronic Medical Records (EMR): Digital storage of patient contact information and basic treatment history.
- Inventory Tracking: Manual entry systems for monitoring dental consumables and supplies.
- Financial Reporting: Basic modules for generating daily revenue reports and invoice printing.
- Static Appointment Calendars: Manual drag-and-drop grids for scheduling that require constant human monitoring.

Advantages:

- Operational Familiarity: Requires minimal training for staff accustomed to basic computer operations.
- Low Entry Cost: Generally affordable for small-scale, single-chair neighborhood clinics.
- Localized Control: Data is stored on-site, providing owners with a sense of direct physical data ownership.

Disadvantages:

- Lack of Diagnostic Intelligence: No AI integration to assist in X-ray analysis or treatment planning, leaving the burden of diagnostic accuracy entirely on human interpretation.
- Fragmented Patient Experience: No patient-facing interface or teledentistry portal, forcing all interactions to be conducted in-person or via unsecured messaging apps.
- Zero Predictive Capability: The systems cannot analyze patient trends to predict future "no-shows" or suggest preventive care interventions, resulting in reactive rather than proactive healthcare.

3.2.5 Conclusion

The S.M.I.L.E platform represents a transformative leap in the digitalization of the Vietnamese dental healthcare sector, moving beyond the limitations of traditional, fragmented management systems toward a unified, intelligent ecosystem. The project effectively mitigates the systemic issues of information asymmetry, clinical inefficiency, and patient anxiety that currently hinder the industry. As Vietnam solidifies its position as a regional hub for dental tourism, the implementation of such a sophisticated technological framework provides the necessary infrastructure to handle complex workflows and international standards. Ultimately, S.M.I.L.E does not merely optimize administrative tasks; it establishes a new benchmark for clinical integrity and patient-centric care, serving as a scalable catalyst for sustainable innovation across the broader ASEAN healthcare landscape.

4. Business Opportunity

The Vietnamese dental industry is currently at a critical digital inflection point. While an expanding middle class and the rise of dental tourism have surged demand for sophisticated restorative procedures, the sector remains hindered by a fragmented technological landscape. Most contemporary platforms rely on rudimentary, manual systems that fail to address "dental anxiety" or provide transparency in pricing and treatment outcomes. For clinic operators, this lack of integration results in significant underutilization of clinical capacity and a deficit in actionable patient data. S.M.I.L.E (Smart Management & Intelligent Life Ecosystem) is strategically engineered to resolve these systemic failures by harmonizing the divergent needs of healthcare consumers and providers within a single, intelligent ecosystem.

Built upon a robust technological foundation of Hyperledger Fabric blockchain and DentalMultiTaskNet AI, S.M.I.L.E transcends traditional booking tools to offer a comprehensive digital healthcare environment. The platform's dual-value proposition delivers unprecedented transparency for patients through immutable medical records and real-time treatment visualizations, while simultaneously empowering practitioners with advanced analytics and automated resource allocation. By leveraging a microservices architecture and IPFS decentralized storage for high-resolution medical imaging, the system ensures both data integrity and the scalability required for high-volume clinical environments.

Furthermore, S.M.I.L.E is uniquely positioned to capitalize on Vietnam's emergence as a regional hub for dental tourism. By integrating localized payment gateways like VNPay and PayOS with comprehensive multilingual support, the platform serves as a strategic bridge for international patients seeking cost-effective, world-class treatments in urban centers like Ho Chi Minh City and Hanoi. This alignment with Vietnam's national digital health strategy—prioritizing telehealth and medical record digitalization—positions S.M.I.L.E not merely as a software product, but as essential infrastructure for the future of dentistry. Ultimately, the platform acts as a catalyst for modernization, establishing a sustainable framework for expansion into the broader ASEAN healthcare market.

5. Software Product Vision

S.M.I.L.E envisions a future where blockchain technology and artificial intelligence seamlessly transform dental healthcare - from diagnosis to treatment management. The core vision is to build a secure, intelligent, and patient-centric platform that redefines how individuals access dental care while empowering practitioners to deliver exceptional treatment in a digitally-driven healthcare economy.

At its heart, S.M.I.L.E bridges the gap between modern patient expectations and dental practice complexities through immutable data integrity and AI-powered diagnostics. The platform provides every patient with a transparent, personalized dental care experience through AI-assisted consultations, blockchain-verified medical records, and teledentistry services that make dental care more accessible and trustworthy.

For dental practitioners, the vision encompasses sophisticated clinical tools including DentalMultiTaskNet AI for X-ray analysis, streamlined appointment management, blockchain-verified patient records ensuring data integrity, and comprehensive analytics that inform clinical decisions and operational efficiency.

Unlike conventional dental management platforms focused on basic scheduling and billing, S.M.I.L.E differentiates itself through blockchain-first architecture, specialized AI diagnostic capabilities, and unwavering commitment to data security. The platform leverages Hyperledger Fabric for immutable records, integrates with Vietnamese payment systems like VNPay, and supports the country's digital health transformation while establishing regional healthcare technology leadership.

Ultimately, S.M.I.L.E's product vision is to become the definitive digital infrastructure for dental healthcare innovation, where patients and practitioners are empowered through verifiable trust, intelligent insights, and seamless care experiences. The platform aspires to set new standards for healthcare data integrity and patient engagement while fostering a transparent, efficient dental healthcare ecosystem that serves as a model for digital health transformation across Southeast Asia.

6. Project Scope & Limitations

The S.M.I.L.E project focuses on being a blockchain-secured, AI-enhanced dental practice management platform, positioning itself as an innovative healthcare technology solution that revolutionizes dental care delivery and patient data management. The scope of this project centers around utilizing cutting-edge blockchain technology and artificial intelligence to deliver core functionalities that enhance clinical accuracy, data integrity, and operational efficiency. Key features include AI-powered diagnostic assistance through DentalMultiTaskNet for X-ray analysis, blockchain-verified medical records using Hyperledger Fabric, intelligent appointment scheduling systems, teledentistry consultation capabilities, and automated treatment plan generation. These advanced functionalities aim to redefine how patients and practitioners interact within the dental healthcare ecosystem, providing unprecedented transparency and clinical support.

While the project emphasizes blockchain security and AI-driven clinical assistance, it will not include features like complex dental laboratory management systems, advanced dental equipment integration beyond imaging devices, or comprehensive hospital information system (HIS) connectivity, as these are beyond the current focus on core dental practice digitalization. The platform also excludes real-time emergency dental services coordination and advanced pharmaceutical inventory management, concentrating instead on the fundamental transformation of dental record keeping and diagnostic assistance.

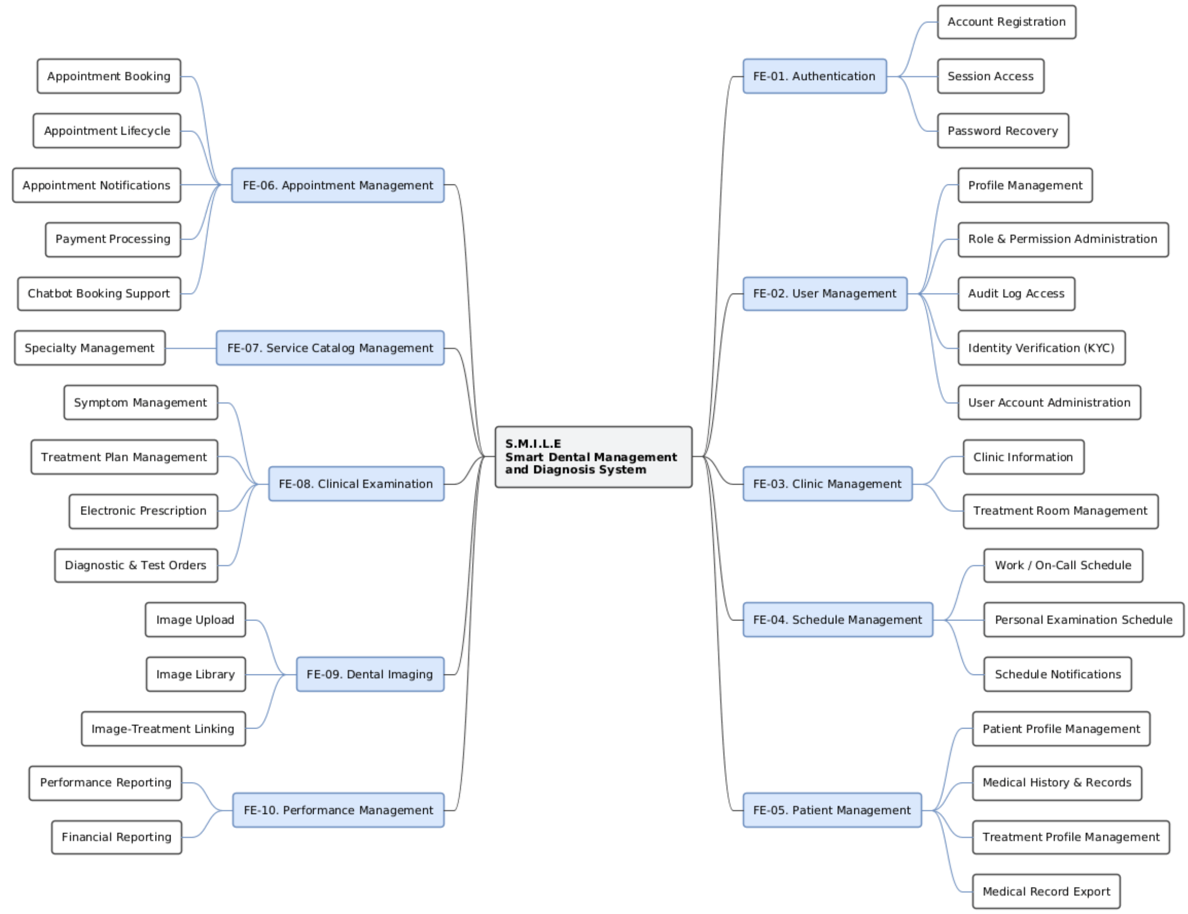
By focusing on blockchain-secured data integrity and AI-enhanced diagnostics as the backbone of its operations, the project aligns with the goal of establishing S.M.I.L.E as a leader in secure, intelligent dental healthcare platforms. This defined scope ensures clear priorities around patient data protection and clinical decision support, realistic expectations for implementation timelines, and a solid foundation for evaluating future enhancements as the platform evolves to serve the broader Southeast Asian dental healthcare

6.1 Major Features

The S.M.I.L.E platform offers a comprehensive set of features designed to enhance the dental healthcare experience for patients and streamline clinical operations for dental practitioners. These functionalities are organized into modular components, ensuring scalability, maintainability, and user-centric interaction while maintaining the highest standards of data security through blockchain technology. The major features of the system include:

- FE-01. Authentication & Authorization**:** Provides secure user identity management with role-based access control (RBAC) via JWT tokens, including account registration through email or Google OAuth, login, logout, and password recovery. This ensures patient data privacy and controlled access to clinical features based on role permissions.
- FE-02. Blockchain Record Verification**:** Enables immutable medical record integrity through Hyperledger Fabric blockchain technology, allowing patients and practitioners to verify the authenticity and completeness of dental records stored in the system.
- FE-03. AI-Powered Diagnostic Assistance: Supports automated analysis of panoramic and cephalometric X-rays using DentalMultiTaskNet AI model, providing preliminary diagnostic insights including tooth segmentation, landmark detection, and pathology identification to assist clinical decision-making.
- FE-04. Appointment Scheduling & Management: Allows patients to book, reschedule, or cancel appointments while enabling clinic staff to manage schedules, view available time slots, and optimize chair utilization through intelligent scheduling algorithms.
- FE-05. Payment Processing Integration: Integrates secure payment methods including VnPay for Vietnamese market, supporting real-time payment processing, refunds, and transaction verification with IP validation and checksum security measures.
- FE-06. Digital Treatment Planning: Provides tools for creating, visualizing, and managing comprehensive treatment plans with automated plan generation based on AI diagnostic insights and clinical assessment data.
- FE-07. Patient Health Record Management: Allows patients to access their complete dental history, view treatment outcomes, track preventive care schedules, and manage medical consent forms through a secure, user-friendly interface.
- FE-08. Clinical Examination Management: Supports dental practitioners in creating examination sessions, writing digital prescriptions, documenting clinical findings, and integrating AI analysis results into patient records.
- FE-09. Multi-Clinic Network Support: Enables management of multiple clinic locations with centralized patient data while maintaining local operational autonomy for scheduling, billing, and clinical workflows.
- FE-10. Staff & Role Management: Allows administrators to manage dental staff accounts, assign role-based permissions, track clinical activities, and maintain professional credentials and certifications.
- FE-11. Patient Communication System: Empowers clinics to send automated appointment reminders, preventive care notifications, treatment follow-ups, and educational content to improve patient engagement and care continuity.Each of these features contributes to S.M.I.L.E's mission of delivering a secure, intelligent, and scalable platform that benefits both patients and dental healthcare providers. The system's blockchain-first architecture ensures

data integrity and patient privacy while AI-enhanced capabilities support clinical excellence and operational efficiency. This modular design allows the platform to evolve with advancing medical technologies and expand into broader Southeast Asian healthcare



6.2 Limitations & Exclusions

- LI-01. Limited AI Diagnostic Scope: AI-powered diagnostic assistance is restricted to panoramic X-rays and 7 common dental conditions only. Advanced 3D imaging analysis and rare pathology detection are not supported.
- LI-02. Vietnamese Payment Methods Only: Payment integration limited to VNPay, MoMo, and ZaloPay. International payment methods and cryptocurrency payments are not supported.
- LI-03. Basic Analytics Dashboard: Analytics limited to standard reporting on appointments and payments. Advanced predictive analytics and machine learning insights are not supported.
- LI-04. Limited Multi-Language Support: System supports Vietnamese and English only. Additional languages and real-time translation services are not available.
- LI-05. No Legacy System Migration: Platform does not include automated migration utilities from existing dental management systems. Data migration must be handled manually.
- LI-06. Basic Notification Channels: Patient communication limited to email, SMS, and push notifications. Advanced personalized content generation is not supported.